

The Rise of Generative AI Increases Job Insecurity and Behavioral Adaptation Traps: Structural Evidence from Heterogeneous Volatility and SVAR Models

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Abstract. This paper constructs a structural macroeconomic framework to measure labor market frictions induced by Generative Artificial Intelligence (GenAI). Unlike previous physical automation, GenAI poses a direct structural shock to non-routine cognitive occupations. Exploiting high-frequency weekly data from the US and Vietnam, we estimate the transmission mechanism from AI-induced technological shocks to aggregate job insecurity and subsequent "behavioral adaptation traps" (e.g., workforce burnout and quitting). Addressing methodological flaws in standard approaches, our pipeline integrates heterogeneous volatility filtering (IGARCH for persistent shocks in the US and EWMA for non-ARCH environments in Vietnam) with a structural vector autoregressive (SVAR) model incorporating psychological rumination lags. Furthermore, accounting for endogenous structural breaks, we utilize split-sample interaction models and difference-in-VAR specifications where long-run cointegration fails. The evidence reveals a severe behavioral adaptation trap in the US, where continuous adaptation to GenAI sharply accelerated aggregate burnout post-ChatGPT. Conversely, Vietnam exhibited an acute initial shock that dissipated due to institutional buffers. These findings highlight the critical need for targeted, technology-mature labor policies over passive observation.

Keywords: Generative AI · Job Insecurity · SVAR · IGARCH · Structural Breaks · Behavioral Adaptation

1 Introduction

Technological innovations routinely disrupt labor markets, historically manifesting as physical automation that displaces routine blue-collar employment. However, the commercialization of Generative Artificial Intelligence (GenAI)—most notably catalyzed by the release of Large Language Models (LLMs) in late 2022—represents an unprecedented macroeconomic shock. By demonstrating capabilities in coding, analysis, and strategic reasoning, GenAI introduces a structural threat to highly skilled, knowledge-intensive professions.

This paper uses a rigorous structural time-series approach to map the causal transmission of GenAI shocks into psychological labor market outcomes. While existing literature extensively documents physical automation’s impact on maladaptive workplace behaviors [2], empirical evidence on the macro-psychological toll of cognitive automation—often conceptualized as “Behavioral Adaptation Traps”—remains sparse. Drawing on Cognitive Appraisal Theory [3], we posit that knowledge workers engage in intensive proactive adaptation when appraising GenAI as an occupational threat. When technological diffusion outpaces human cognitive plasticity, these efforts systematically backfire, driving aggregate workforce burnout.

We contribute to the literature by addressing severe methodological limitations inherent in high-frequency time-series analysis of behavioral data. First, we abandon “one-size-fits-all” variance modeling by applying appropriate heterogeneous volatility filters. Second, we relax unrealistic recursive Cholesky assumptions by imposing structural delays in our SVAR [1], reflecting realistic psychological “rumination lags.” Third, we evaluate structural asymmetry between the US and Vietnam post-ChatGPT, explicitly accounting for cointegration breakdowns and unit root break biases.

2 Conceptual Framework and Hypotheses

The transmission of technological shocks to labor outcomes operates through cognitive appraisal [3]. GenAI replicates non-routine cognitive functions, directly undermining the white-collar competitive advantage and initiating structural labor market anxiety.

Hypothesis 1 (H1): *Macro-level Generative AI shocks induce significant short-to-medium-run structural shifts in aggregate job insecurity.*

In response to structural threats, employees engage in proactive adaptation. However, expending excessive cognitive resources to match algorithmic velocity causes resource depletion—the *Behavioral Adaptation Trap*.

Hypothesis 2 (H2): *Generative AI shocks structurally drive aggregate workforce burnout and job quitting behaviors, subject to a psychological rumination lag.*

The transmission of these shocks is nonlinear. The widespread accessibility of ChatGPT arguably served as a major structural break, with institutional heterogeneity dictating different adjustment paths across advanced (US) and emerging (Vietnam) markets.

Hypothesis 3 (H3): *The transmission mechanism is structurally altered post-ChatGPT, displaying severe adaptation traps in the US and dampened adjustment in Vietnam.*

3 Empirical Strategy

To rigorously trace structural causal paths and avoid spurious high-frequency correlations, we employ a three-stage methodology.

3.1 Heterogeneous Volatility Filtering: IGARCH and EWMA

A preliminary ARCH-LM test reveals severe structural heterogeneity in volatility. In the US, AI volatility shocks exhibit long-memory persistence ($\alpha + \beta \approx 1$), necessitating an Integrated GARCH, $IGARCH(1,1)$, specification to prevent unstable statistical inference:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + (1 - \alpha) \sigma_{t-1}^2 \quad (1)$$

Conversely, Vietnamese AI shock data exhibit no conditional heteroskedasticity (no ARCH effects). Imposing GARCH here would be theoretically invalid. Thus, we extract structural uncertainty using an Exponentially Weighted Moving Average (EWMA) proxy. The filtered variance serves as our macroeconomic AI Volatility measure (AI_{Vol}).

3.2 SVAR with Psychological Structural Delays

Let $X_t = [AI_{Vol}, GenAI_{Shock}, Job_{Insecurity}, Burnout, Job_{Quitting}]'$. Standard recursive Cholesky decompositions assume sequential behavioral reactions within a single week, which violates psychological realities. We identify the structural model [1]:

$$A_0 X_t = c + \sum_{i=1}^p A_i X_{t-i} + \varepsilon_t \quad (2)$$

by replacing the lower-triangular A_0 with theoretically driven zero-restrictions. Specifically, we restrict contemporaneous (same-week) shocks from AI_{Vol} to $Burnout$ and $Job_{Quitting}$ to zero, enforcing a minimum lag (e.g., $t - 1$ to $t - 4$) to account for the cognitive processing and rumination required before a worker burns out or resigns.

3.3 Cointegration Diagnostics and Regime-Dependent ARDL

Standard Augmented Dickey-Fuller (ADF) tests are biased toward non-stationarity under structural breaks. We verify integration orders using break-robust unit root tests (e.g., Zivot-Andrews). Where bounds testing [4] confirms stable long-run cointegration, we estimate an Error Correction Model (ECM). Crucially, for periods where cointegration explicitly fails (e.g., US Job Insecurity pre-break), we abandon the ECM and estimate a *VAR in differences* to map short-run structural transmission without forcing a non-existent long-run equilibrium:

$$Y_t = \alpha + \beta_{base} G_t + \beta_{inter} (G_t \times D_t) + \gamma D_t + \Omega Z_t + v_t \quad (3)$$

where D_t indicates the post-ChatGPT period, and β_{inter} captures the slope shift.

4 Empirical Results

4.1 Diagnostics and Cointegration Breakdown

Standard ADF tests yielded "Break Bias Flags" for critical US variables, confirming the necessity of our structural-break framework. ARDL bounds testing [4] revealed that cointegration is highly regime-dependent. Specifically, US models for Job Quitting post-break lack valid cointegrating vectors, rendering long-run equilibrium forecasts invalid. Consequently, our findings emphasize explanatory structural shifts rather than predictive accuracy, avoiding the pitfalls of naive forecasting benchmarks.

4.2 Time-Varying Sensitivities and Behavioral Traps

Table 1 reports parameter estimates capturing structural breaks in the transmission mechanism.

Table 1: Estimates of Structural Breaks in the Transmission Mechanism

Country	Dependent Variable	β_{base}	β_{post}	β_{inter}	p -val	Level Shift (γ)
US	Job Insecurity	0.59596	0.82044	0.22448	0.4038	1.04717***
	Burnout	-0.01581	0.59023	0.60604***	0.0000	-0.03473
	Job Quitting [†]	0.16508	0.02539	-0.13969	0.3806	-0.11424***
VN	Burnout ($lMDI$)	2.19009	0.47258	-1.71752***	0.0000	-0.05621*

Note: ***, **, and * denote significance at 1%, 5%, and 10%. β_{inter} captures the post-ChatGPT shift. [†]Indicates parameters derived from short-run dynamics due to lack of robust cointegration.

The empirical evidence confirms a profound divergence. In the US, the baseline sensitivity of workforce burnout to GenAI shocks was negligible ($\beta_{base} = -0.01581$). However, post-ChatGPT, this sensitivity accelerated sharply ($\beta_{inter} = 0.60604, p < 0.01$). This dynamic exposes a severe behavioral adaptation trap: continuous proactive adaptation efforts in the highly fluid US labor market systematically amplified aggregate burnout [2]. We also observe significant permanent level shifts ($\gamma = 1.04717$) in US Job Insecurity, confirming an immediate structural macroeconomic anxiety shock.

Conversely, Vietnam exhibited a dampening trajectory. The initial sensitivity of Burnout was exceptionally high ($\beta_{base} = 2.19009$), reflecting acute friction. Post-break, the interaction term aggressively corrected this ($-1.71752, p < 0.01$). This confirms that institutional buffers and gradual technological assimilation in the emerging market helped dissipate initial panic, sidestepping the chronic adaptation trap seen in the US.

5 Conclusion

This paper reconstructs the macroeconomic transmission of Generative AI shocks into labor market frictions, stripping away the biases of standard modeling. By integrating heterogeneous volatility (IGARCH/EWMA) and structurally delayed SVARs, we provide robust evidence that the public diffusion of GenAI activated severe behavioral adaptation traps in advanced economies (the US), while manifesting as transient acute shocks in emerging markets (Vietnam). Passive observation of AI integration is insufficient; policymakers must deploy targeted psychological and institutional buffers to navigate the algorithmic transition.

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